

A diagram about funcoids and reloids

OPG-60024	L1 - Tractable	Partiellement résolu	Topology
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Énoncé original (anglais, reproduction mot pour mot)

Define for posets with order $\sqsubseteq$:

- $\Phi_{\ast} f = \lambda b \in \mathfrak{B}: \bigcup \{ x \in \mathfrak{A} \mid f x \sqsubseteq b \}$;

- $\Phi^{\ast} f = \lambda b \in \mathfrak{A}: \bigcap \{ x \in \mathfrak{B} \mid f x \sqsupseteq b \}$.

Note that the above is a generalization of monotone Galois connections (with $\max$ and $\min$ replaced with suprema and infima).

Then we have the following diagram:

What is at the node "other" in the diagram is unknown.

Conjecture "Other" is $\lambda f \in \mathbf{FCD}: \top$.

Question What repeated applying of $\Phi_{\ast}$ and $\Phi^{\ast}$ to "other" leads to?

Particularly, does repeated applying $\Phi_{\ast}$ and/or $\Phi^{\ast}$ to the node "other" lead to finite or infinite sets?

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